

FURKAN BODUR

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Legally Authorized to Work in Poland (Spouse of a Polish Citizen) • No Work Sponsorship Required

AI Software Engineer with 9 years of professional engineering experience, specializing in production-grade Generative AI orchestration, custom Model Context Protocol (MCP) systems, and enterprise Retrieval-Augmented Generation (RAG) pipelines over the past 2 years. Combines a deep background in mission-critical software validation, test automation, and CI/CD infrastructure with LLM application design.

TECHNICAL SKILLS MATRIX

Artificial Intelligence	Multi-Agent Systems, RAG Pipelines, Vector Search (pgvector), Custom Agentic Frameworks, Model Context Protocol (MCP), LLM Evaluation & Validation, Semantic Memory Layers, Response Streaming (SSE), Vision-Language Models (VLMs), Embedding & Reranking
Languages & Core Dev	Python, Node.js, TypeScript, SQL (PostgreSQL), RESTful APIs, Asynchronous Programming, HTTP
DevOps & Infra	Google Cloud Run, AWS EC2, Docker, Siemens-Cleared Containers, CI/CD Pipelines (GitLab CI, Jenkins), Linux Systems
Software Quality	Automation Engineering, Core Framework Architecture, Full-Stack Validation Pipelines, Playwright, Selenium

PROFESSIONAL EXPERIENCE

Senior Software Development Engineer in Test (SDET) — SIEMENS Foundational Technologies *Feb 2022 – Present*

Focus: GenAI Tooling, RAG Architecture & Workflow Automation

- Designed and developed a production-grade project knowledge base tool utilizing Siemens-hosted internal LLM models, implementing advanced embedding and reranking pipelines to provide highly accurate, enterprise-governed semantic search.
- Architected and deployed a custom multi-agent Python framework consisting of 5 autonomous agents to automate complex internal engineering workflows, accelerating manual processing speeds by 80%.
- Engineered secure Model Context Protocol (MCP) servers using Node.js and TypeScript to safely interface language models with internal Siemens Jira and GitLab APIs for automated issue tracking.
- Managed full-stack Python pipelines and internal AWS cloud environments, deploying software via Siemens-cleared containers integrated with Jenkins and GitLab CI for continuous delivery.
- Optimized application latency and user experience by designing response-streaming architectures via Server-Sent Events (SSE) for real-time text output.
- Implemented a dual-layer LLM memory topology combining client-side browser caching with long-term semantic vector persistence to maintain context continuity across complex user sessions.

Software Test Engineer & UAV Pilot — TURKISH AEROSPACE INDUSTRIES (TAI / TUSAŞ) *Jun 2017 – Feb 2022*

- Led software verification, functional validation, and automated testing for mission-critical Unmanned Aerial Vehicle (UAV) flight-control and ground systems, ensuring 100% adherence to strict safety and regulatory requirements.
- Collaborated closely with cross-functional core engineering teams to isolate, reproduce, and patch complex system defects, scaling automated test scripts to reduce overall regression cycles.

SELECTED AI PROJECTS & RESEARCH

Academic Research Paper — Lead Author *AICCONF 2026 | Prague*

"Bridging the Gap: An Agentic AI Framework for Reverse-Engineering Automated Test Scripts into Natural Language Requirements"

4th Cognitive Models and Artificial Intelligence Conference — Comparative analysis of 3 LLM models over 300+ industrial Python automation scripts.

Telegram-Integrated Multi-Agent Pipeline — Creator *Google Vertex AI & Cloud Run*

Built a serverless, containerized image generation pipeline on Google Cloud Run utilizing Google's Agent Development Kit (ADK). Constructed stateful, graph-based agent workflows connecting a custom Telegram bot to Gemini models and image upscaling microservices.

Digital Twin Conversational Knowledge Base — Creator *Google AI Studio & MCP*

Developed a web application leveraging Google AI Studio to transcribe, index, and query multimedia enterprise knowledge assets. Implemented custom stateful Memory Banks and MCP (Model Context Protocol) concepts to safely interface underlying data tools with language models.

EDUCATION

Bachelor of Science in Computer Engineering *Aug 2013 – Jun 2017*

Gazi University — Ankara, Turkey